Resolution Geometry

Rigidity: Why Resolution Geometry Is Not a Choice

Universal Properties, Import Rigidity, Hypothesis Necessity, and the Anti-Ladder

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

October 2026

Abstract

A dependency-ordered framework invites a familiar objection. Its imported theorems could have been others, its hypotheses could have been chosen differently, the gaps between its rungs could hold other structure, and its order is a presentation. In short, the objection says the framework is one choice among many.

This paper answers the objection for Resolution Geometry (RG) by separating four kinds of rigidity and measuring each on the print edition of the ladder:

1. *Universal-property rigidity.* The quotient, the effective symmetry group, natural descent, specialization maps and the panel antichain are each characterized by a universal property, hence unique up to unique isomorphism.
2. *A dichotomy.* Every structure in an RG object is either law-natural, and therefore forced by the kernel and the stabilizer, or declared, and therefore named. There is no third source, so the "space between rungs" is exactly the set of named declarations.
3. *Hypothesis necessity.* Each of the sixteen named hypotheses has a computed countermodel in which its rung fails. None is decorative.
4. *Import rigidity.* Each of the twenty-six imported rungs is a uniqueness theorem for a requirement already imposed earlier. Its residual freedom is a finite parameter fixed by a named declaration.

The dependency relation (the *anti-ladder*) is a directed acyclic graph with a single source, L0, and height 23, and the rung numbering is one of its linear extensions. We report honestly what is *not* forced: the numbering (median positional slack 93), the grouping into strata, the proof routes, and the declarations themselves. A six-check battery passes 6/6 on five seeds, and three canaries are caught.

Contents

1 The objections, stated sharply **1**

2 The dichotomy: natural or declared **2**

3 Universal-property rigidity **2**

4 Import rigidity **3**

5 Hypothesis necessity **4**

6 The anti-ladder: what the order is and is not **5**

7 What is not forced **6**

8 Anticipated objections **7**

9 Conclusion **7**

A Reproducibility **7**

1 The objections, stated sharply

O1 *Imports.* The imported theorems (Čencov, Gleason, Malament, Oseledets, ...) could be replaced by others, and the geometry would change.

O2 *Gaps.* Between two rungs, additional structure could be inserted, so the ladder is not tight.

O3 *Hypotheses.* The named hypotheses are modelling choices; weaker or different ones might serve.

O4 *Order.* The position of a rung is an editorial decision.

O5 *Framework.* Another formalism (category theory, information geometry, Bayesian inference) would give a different theory of the same phenomena.

Each objection is answered by a different kind of rigidity, and each answer is tested by computation (UM01–UM06). Where the answer is "this is not forced", we say so (Section 7).

2 The dichotomy: natural or declared

Definition 2.1. A structure on an RG object is *natural* if it is a pullback $F^*\mu$ of a structure μ on law space that is invariant under the declared relabelings. It is *declared* if it depends on data beyond the joint law: a protocol, task, admissible class, nuisance convention, anchor, latent covariance, or a stated subgroup of Γ.

Theorem 2.2 (Dichotomy)**.** *Every structure appearing in an RG statement is natural or declared. Natural structures are forced: they are basic for $\mathcal{K}$, they descend to the quotient, and they are Γ-invariant (ladder L19–L20). Declared structures enter only through named hypotheses, which are listed in the paper and in each rung's status.*

Proof. A structure either depends on the observation system only through the joint law F or it does not. In the first case it is a function of law values, hence a pullback from law space. Invariance under relabeling is required because relabeled laws describe the same experiment. Natural descent then makes it forced. In the second case it uses information not contained in F, which by definition is a declaration; the ladder names every declaration it uses (UM06). □

Corollary 2.3 (No hidden gap; answer to O2)**.** *Any structure that could be "inserted between rungs" is either natural, in which case it is already determined by $\mathcal{K}$ and Γ and adds no freedom, or declared, in which case it must appear as a named hypothesis. The ladder is tight in this precise sense: there is no third kind of input.*

The battery parses the print edition and confirms that every one of the 116 rungs carries an admissible status, and that every hypothesis in a status is defined in the paper (UM06).

3 Universal-property rigidity

Theorem 3.1 (Forced objects are unique)**.** *The following are characterized by universal properties, hence unique up to unique isomorphism:*

1. *the quotient Q, the coequalizer of the kernel pair: the coarsest space through which F factors;*

2. *the joint kernel, the meet of protocol kernels; and garbling, an inclusion of kernels, functorial in the garbling;*

3. Γ, *the image of* Stab *in the automorphisms of* Q*;*

4. *task descent:* $\bar{T}$ *is the unique map with* $T = \bar{T} \circ \pi$*;*

5. *specialization: the unique map* $Q_\varepsilon \to Q_0$ *factoring* π_0 *when* $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$*;*

6. *the minimal-panel antichain, the unique generating antichain of an upward-closed family;*

7. *the efficient metric, the information of the profiled experiment (Schur complement);*

8. *the canonical connection under a declared covariance: the unique minimal-norm lift.*

Why a groupoid (answer to part of O5). Equality of laws is an equivalence relation, and equivalence relations are exactly the principal groupoids with trivial isotropy. The groupoid is therefore not a stylistic choice; it is what equality of laws is. Any reformulation, categorical or otherwise, must recover the same coequalizer and is isomorphic to it [8].

Test (UM01). We take all 52 partitions of a five-point latent set and 120 random laws. In every case the kernel partition is the unique partition that both factors the law and is coarsest such. Every other candidate fails one of the two properties. A canary that substitutes the discrete partition is caught.

4 Import rigidity

Every imported rung answers a requirement that the ladder has already imposed. The table states:

- the requirement and the characterization theorem that solves it;
- what an alternative would violate;
- the residual freedom left by the theorem, and the declaration that fixes it.

The listed choices are the only ones consistent with the earlier rungs.

rung	requirement already imposed	characterization (uniqueness)	alternative violates	residual freedom → declaration
L28	law-natural metric, invariant under Markov morphisms (L34)	Čencov: multiples of Fisher only	Euclidean, inverse-square, Wasserstein (need a ground metric)	scale → anchor (L97, L107)
L21, L26	stratified quotient of a regular law	slice theorem; semicontinuity; Hardt finiteness	—	none
L29	reconstruction of second-order data	regular reconstruction	—	none on regular strata
L34	garbling enlarges the kernel (L4)	data-processing inequality	any monotone-increasing metric	none
L37	quantum Markov (CPTP) invariance	Petz: operator-monotone f	non-operator-monotone f	$f \to$ task (L38), identified (L39)
L41	noncontextual probability on quantum events	Gleason, Busch: trace rule	$\mathrm{Tr}(\rho P)^\gamma$, $\gamma \neq 1$ (normalization)	none
L42	joint law of protocols (L3)	Fine: coupled law iff local	assuming a coupled law	none

rung	requirement already imposed	characterization (uniqueness)	alternative violates	residual freedom → declaration
L45	finite data on a boundary	finite-sample reconstruction	—	sample size → certificate
L47, L49	Bayes with fibre-constant likelihood	Bernstein–von Mises; LAN	uniform prior washing	prior on fibres → declared
L54	certification from samples	Le Cam two-point bounds; DKW	sample-based TV locks	class → H-adm
L56	laws without a common dominating measure	pairwise domination; Feldman–Hájek	Fisher on singular directions	none
L65	regular quadratic cone	conformal-class determination	metric from the cone alone	conformal factor → scale (L72)
L67, L69, L70	residual characteristic sets	Hardt / o-minimality; cone duality; Gårding	nonconvex well-posed cones	degree → external (L114)
L74	connection on regular strata	induced Levi-Civita by projection	arbitrary affine connections	none for the observable part
L82	duration memory	Baker–Campbell–Hausdorff	additive memory	none
L84, L85, L87	entropy and exponents of dynamics	Misiurewicz–Szlenk; Ruelle, Pesin; cone-field criterion	any other entropy rate	none
L88, L90	forecast frontier; information cost	Oseledets; Brudno	protocol-dependent slopes	intercept → protocol
L95	concatenation of counted lengths	Cauchy–Hölder: ratio scale	nonlinear scales (non-additive)	unit → SI anchor
L99	isometries of the physical slice	Beckman–Quarles ($n \geq 2$)	non-isometric unit-preserving maps	none
L102	radar distance on stationary slices	Landau–Lifshitz spatial metric	static metric with rotation	none (non-stationary: no-go L104)

Theorem 4.1 (Import rigidity)**.** *For every imported rung, either the imported theorem is the unique structure satisfying a requirement already imposed earlier, or it classifies a family whose parameter is fixed by a named declaration. In no case is an imported structure a free choice.*

Proof. By the table. Each entry cites a characterization theorem whose hypothesis is a requirement of an earlier rung. The residual column lists every remaining parameter together with the rung that declares it. □

Tests (UM02, UM03).

- *Markov invariance.* Under random Markov embeddings Fisher and Hellinger (which is Fisher up to the factor $\frac{1}{4}$) are invariant to machine precision. Euclidean changes by 0.50, inverse-square by 1.00.
- *Invariant forms.* The space of invariant symmetric forms has dimension 1 under $SO(3)$ (Schur), 1 under the Lorentz group, 2 under $SO(2)$ (reducible isotropy: anchor budget 2), and 6 without isotropy.
- *Probability rule.* The trace rule normalizes in every basis; $\gamma = \frac{1}{2}$ and $\gamma = 2$ deviate by 0.70 and 0.61.
- *Scales and lifts.* Only linear scales are additive. The minimal-norm lift is unique against 200 alternatives.
- *Petz family.* The SLD, BKM and WY members give 1.000, 1.155 and 1.111 on the same tangent; the family is real, and the declared task fixes the member.

5 Hypothesis necessity

Theorem 5.1 (No decorative hypothesis)**.** *Each named hypothesis has a countermodel in which it fails and the conclusion of its rung fails.*

hypothesis	computed countermodel (UM04)	rung that fails
H-unif	moving bump: pointwise limit, identified pair separates	kernel upper semicontinuity
H-hell	$\mathcal{N}(\varepsilon y, \varepsilon^2)$: Hellinger constant while means converge	distinguishability specialization
H-adm	concentrating densities keep TV $\geq \frac{3}{4}$ after exact transfer	uniform forgetting
H-cov	vertical shear preserves the law, maps any lift to any other	canonical connection
H-SI	dilation (x, c, L) is a null direction of two delays	metre
H-exch	doubling-block record: frequencies oscillate	probability from frequencies
H-task	SLD, BKM and WY differ on one tangent	unique quantum metric
H-med	opposite temperature gradients, equal delays	resolved medium
H-reg	fold: Fisher 0 at the wall, nonzero rescaled jet	tangent sequence
H-nuis	efficient information $\to 0$ at nuisance collinearity	efficient metric continuity
H-coh	completeness holds at even points only	completeness on a stratum
H-sound	unsound arrow merges different laws	quotient comparison
H-int	scale cocycle with product 1.25: residual 0.13	global scale representative
H-stat	5% drift: transferred scale off by 4.3%	ergodic transfer
H-Γ	rotations without boosts: 2 invariant forms on $1+3$	Lorentz forcing
H-val	nonlinear detector passes its anchor, fails at 15 m	anchor identification

A canary that makes the three Petz members coincide is caught. The countermodels are minimal: each violates only the hypothesis it targets in a model where the rung's other inputs hold or are vacuous.

6 The anti-ladder: what the order is and is not

Theorem 6.1 (Order rigidity)**.** *The dependency relation of the ladder is a partial order with a unique minimal element L0. Every rung is a descendant of L0, and the rung numbering is a linear extension.*

Measured on the print edition:

- 116 rungs, 297 dependencies, 192 of them essential;
- height 23, width 46.

The longest chain is

$$\mathrm{L0} \to \mathrm{L1} \to \mathrm{L2} \to \mathrm{L18} \to \mathrm{L21} \to \cdots \to \mathrm{L27} \to \mathrm{L83} \to \cdots \to \mathrm{L90} \to \mathrm{L108} \to \cdots \to \mathrm{L113},$$

which runs from equality of laws through strata and specialization to dynamics, the chaos frontier, assembly and closure. A canary injecting a backward dependency (L5 $\leftarrow$ L50) is caught.

Proposition 6.2 (Positions are presentation; answer to O4)**.** *In any linear extension, a rung v occupies a position in $[\#\mathit{ancestors}(v),\ n-1-\#\mathit{descendants}(v)]$. Only L0 has a fixed position, and the median slack is* 93*; imported rungs have slack between* 59 *and* 105*. Counting all linear extensions is #P-complete [7], so they are not enumerated. The numbering is therefore a presentation of the poset. The poset is the claim. Every theorem, and every dependency between theorems, is invariant under renumbering.*

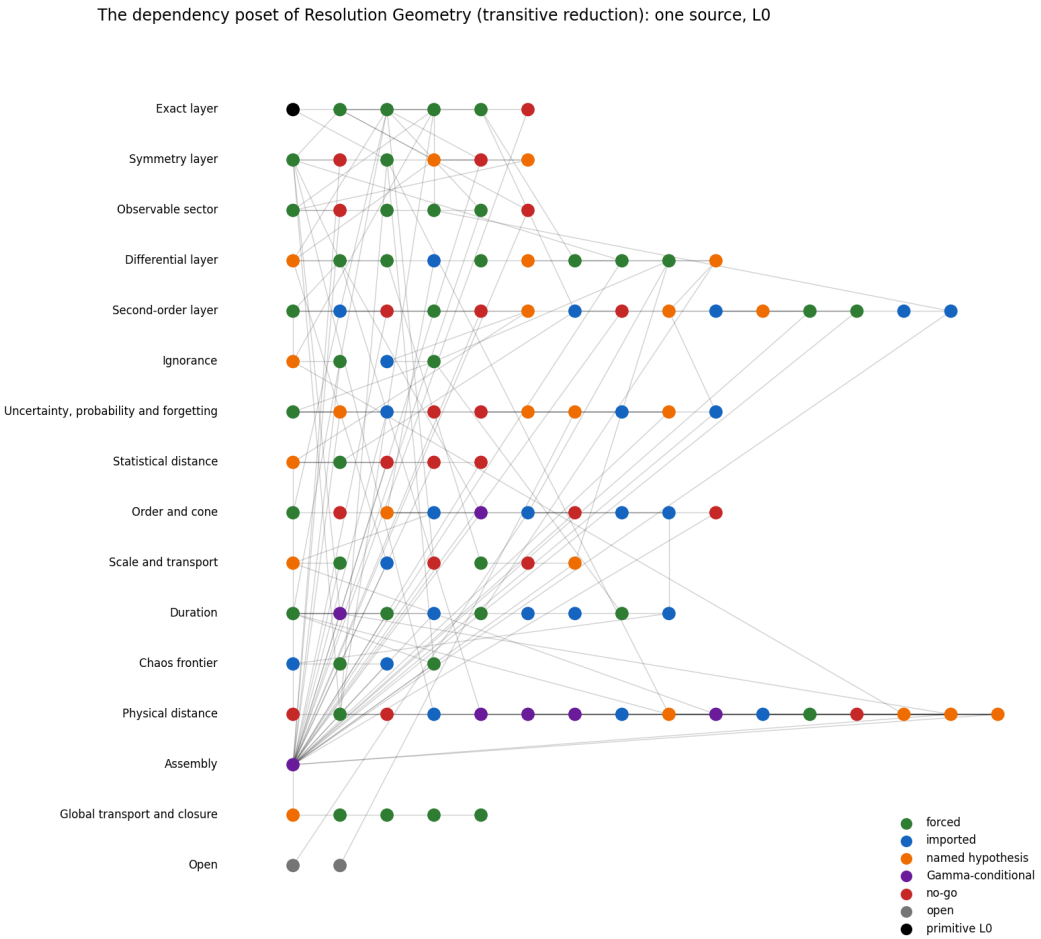


Figure 1: The dependency poset of the print edition (transitive reduction, 192 essential edges), coloured by status. Every rung descends from the single primitive L0, the equality of laws.

Remark 6.3 (Proof edges versus definitional edges). Some edges record the proof route chosen in the paper rather than definitional necessity. A different proof can remove such an edge, but it can never reverse it into a cycle, and it cannot remove the definitional backbone:

- every statement mentions objects defined from L0;
- walls need strata, and strata need the tangent sequence;
- physical distance needs the counted channel.

Shorter proofs make the poset sparser; they do not make RG less forced.

7 What is not forced

1. *Numbering and strata.* These are presentation (Proposition in Section 6).
2. *Proof routes.* These change edges, not theorems.
3. *Declarations.* These comprise the protocol, task, admissible class, anchors, latent covariance, nuisance convention, declared subgroups of Γ, and target precision. They are not derived, and this is a theorem: no task-free equivariant selector exists (ladder L13). They are named and tested.
4. *Normalization constants.* The Fisher scale and the Hellinger factor $\frac{1}{4}$ are fixed by anchors or by convention, and they never change a statement.

Nothing else in the ladder is a choice.

8 Anticipated objections

"Why Fisher and not Wasserstein?" A Wasserstein distance needs a ground metric on outcomes, which is declared structure. When a protocol declares one, as in the Navier–Stokes contraction lock, RG uses it there and says so. As a law-natural metric, Markov invariance leaves only multiples of Fisher.

"Different hypotheses would give a different geometry." Weaker hypotheses fail by the countermodels of Theorem 5.1. Stronger ones prove the same rungs and add declarations, which appear as named inputs.

"The imports are where the content is; RG just collects them." RG claims no new primitive. Its content is the forced assembly:

- a single source;
- universal properties;
- one shared symmetry group (the kernel–stabilizer theorem);
- typed failure;
- each import entering exactly where an earlier requirement calls for its characterization.

Remove any import and a requirement is left without a solution. Replace it, and an earlier rung is violated.

"A categorical or Bayesian formulation would differ." Each object of Theorem 3.1 is defined by a universal property, so any formulation that represents the same experiment represents the same objects up to unique isomorphism. A Bayesian formulation recovers the uncertainty stratum and its dichotomy: priors are forgotten on the quotient and never along the kernel (ladder L47).

9 Conclusion

RG is not one geometry among many for the same declared experiment. Its forced part is characterized by universal properties, and its imports are uniqueness theorems for requirements it has already imposed. Its hypotheses are each necessary, and its order is a single-source poset. What remains free is named: presentation, proof route, and declarations. This answers the five objections with theorems, a dependency graph and computed countermodels, not with canaries alone.

A Reproducibility

`rg_rigidity_attack.py --tex "Resolution Geometry.tex"` passes 6/6 on seeds 20261021, 1, 42, 777 and 2026. `--figs` writes the poset figure, and `--canary` runs three canaries:

- the discrete partition substituted for the kernel;
- coinciding Petz members;
- a backward dependency injected into a copy of the ladder.

All are caught. Corrections before registration:

1. Several hypothesis countermodels were first written as literal arithmetic; they were replaced by computations.

2. The order test first reversed transitive-reduction edges, which cannot create cycles; it was replaced by the positional-slack analysis.

3. The backward-dependency canary first used a regex that missed the row; the injection is now asserted.

References

[1] M. Rada, *Resolution Geometry: Indistinguishability, Its Symmetries, and the Geometry of What Observations Determine*, print edition, 2026.

[2] M. Rada, *The Groupoid of Resolution*, 2026.

[3] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.

[4] D. Petz, Monotone metrics on matrix spaces, *Linear Algebra Appl.* 244 (1996), 81–96.

[5] A. M. Gleason, Measures on the closed subspaces of a Hilbert space, *J. Math. Mech.* 6 (1957), 885–893.

[6] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Phys. Rev. Lett.* 48 (1982), 291–295.

[7] G. Brightwell, P. Winkler, Counting linear extensions, *Order* 8 (1991), 225–242.

[8] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.